

ECSE 4660/6660: Internetworking of Things

Lecture 4: PHY Layer Fundamentals – Part I

Ish Jain
ECSE, RPI

Spring 2026

Administrative

- Use Piazza for Lab questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
 - Asking Questions by email or Webex is not recommended
 - Piazza allows peer-discussion and faster response (can even post anonymously)
 - 5% Class Participation grades
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Project group formation **due 1/30**
 - Add Project Group here [IoT_SP26_project.xlsx](#)
- Project Proposal submission **due 2/6**
 - Template Provided
 - Rubrics Provided
- HW1 to be released this Friday on **LMS** and due in 1 week: **due 2/6**
- Lab is located in JEC 5312
 - Access is provided to all registered students
 - Collect Lab-kit (Rpi etc) during TA hours
 - You can visit lab anytime Mon-Fri 7AM-10PM
 - Labs are done **individually**-Plagiarism penalized heavily (See Syllabus pdf)
 - **TA hours 4-5 PM** daily in 5312 lab
 - Lab 1 released; due **Fri, 1/30**
 - Lab 2 to be released this week; due **Fri, 2/13**

Proposal Template

[A descriptive title]

ECSE 4660/6660 Internetworking of Things

[Date]

[team member1, abcde@rpi.edu, ECSE 4660]

[team member2, abcde@rpi.edu, ECSE 4660]

1. Objective

This project will develop xx and demonstrate yy. xx will help people achieve goal zz. Our idea/ demonstration will benefit society/students/institutes/world because... (Think of this as your next **startup pitch**). This should identify clearly the **research question** you're addressing.

2. Innovation and Literature Review

This is an **innovative idea** and never demonstrated before in this form – some relevant projects found online are listed below... We are different from these works because...



3. Methodology

A clear statement of your **research methods**, i.e., how are you going to solve the problems you've raised and motivated in the previous paragraph?

We will utilize xyz sensors and develop a protocol to communicate the data over abc.

The system diagram is shown in Figure 1. (One or more representative figures or **block diagram** to show your hardware/system flow.)

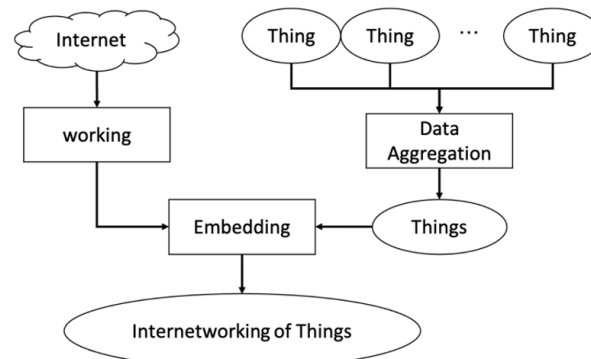


Figure 1 A schematic diagram of XX.

Proposal Template

4. Hardware/Software Used

We will use **compute platform**, e.g., RPi's, such-and-such **sensors**, PC running program V to visualize output, etc. The **program** will be written in Python/C/Java..., and the sensors will be from the IoT lab.

Make sure to check the list of sensors posted on Piazza. If the sensor is not listed and you don't possess them yourself, we might be able to order from Amazon up to certain budget. **Highlight** the need to buy new sensors/hardware in your report.

5. Demo and Deliverable

We will **demonstrate** the use and operation of XX in a **live demo** scenario. The algorithm/protocol will be shown by simulated/real application/or both. Students at 6000 level – describe your

security component here.

6. Plan and Timeline

A statement of plan and schedule, preferably in a table format, to convince us (and yourself!) that you can complete the project by the end of the term.

Actionable Item	Timeline
x	Mar 2 -16 (two weeks)
y	

Lab 1: Linux, Rpi, and Sensors

Part 1 — Linux Basics (VM + SSH/SCP)

- Set up Ubuntu VM (recommended prebuilt VirtualBox image)
- Practice core commands + package installs (apt update/upgrade)
- Install/enable SSH server, and use SSH + SCP
- Publish a small “bio” file to your RPI homepage server

Part 2 — Raspberry Pi + Sensors

- Configure the RPi (monitor setup OR headless SD-card setup)
- Enable SSH, connect from your laptop, transfer files
- Interface sensors (DHT11, HC-SR04) and capture evidence
- Build a simple RFID app (RC-522): write tag → read tag

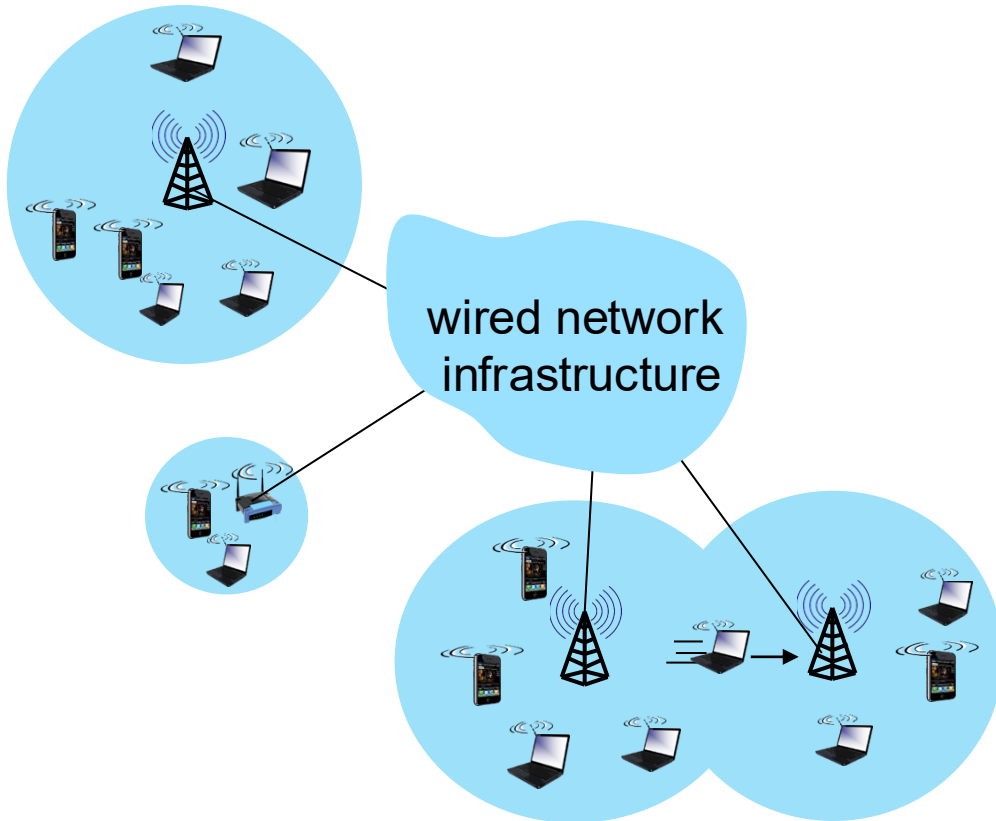
Wireless Overview

1. Overview of Wireless technologies
2. Antenna Design
3. Attenuation
4. Free space loss
5. Noise
6. Multipath & Fading (NLoS)

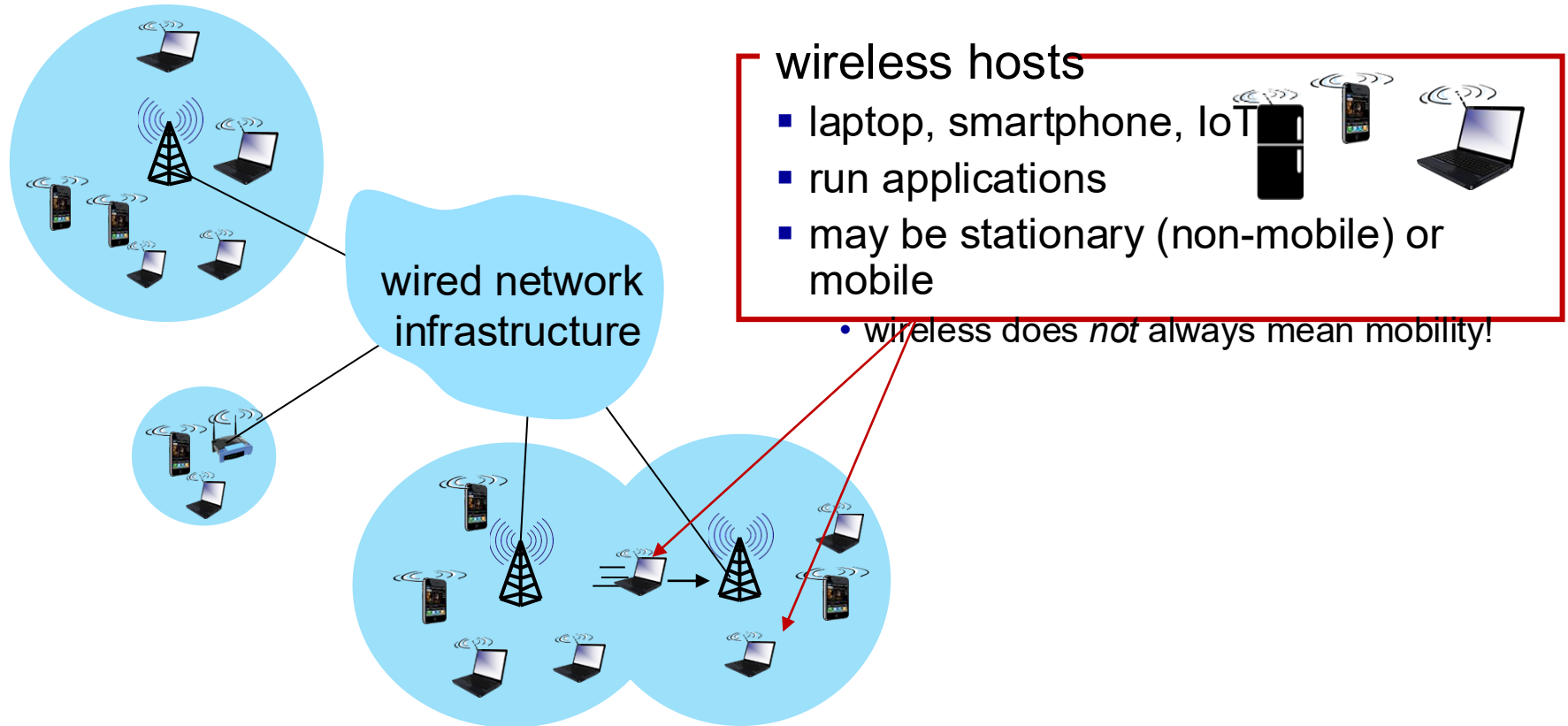
References

- V. Garg, *Wireless Communications & Networking*, Morgan Kauffman, 2007. (available as e-book for RPI students)
 - Chapter 3: Radio Propagation and Path Loss Models
 - Chapter 10: Antennas, Diversity, and Link Analysis
- L. Couch, *Digital and Analog Communication Systems*, Pearson, 2012.
 - Chapter 1: Introduction
 - Chapter 8: Wire and Wireless Communication Applications

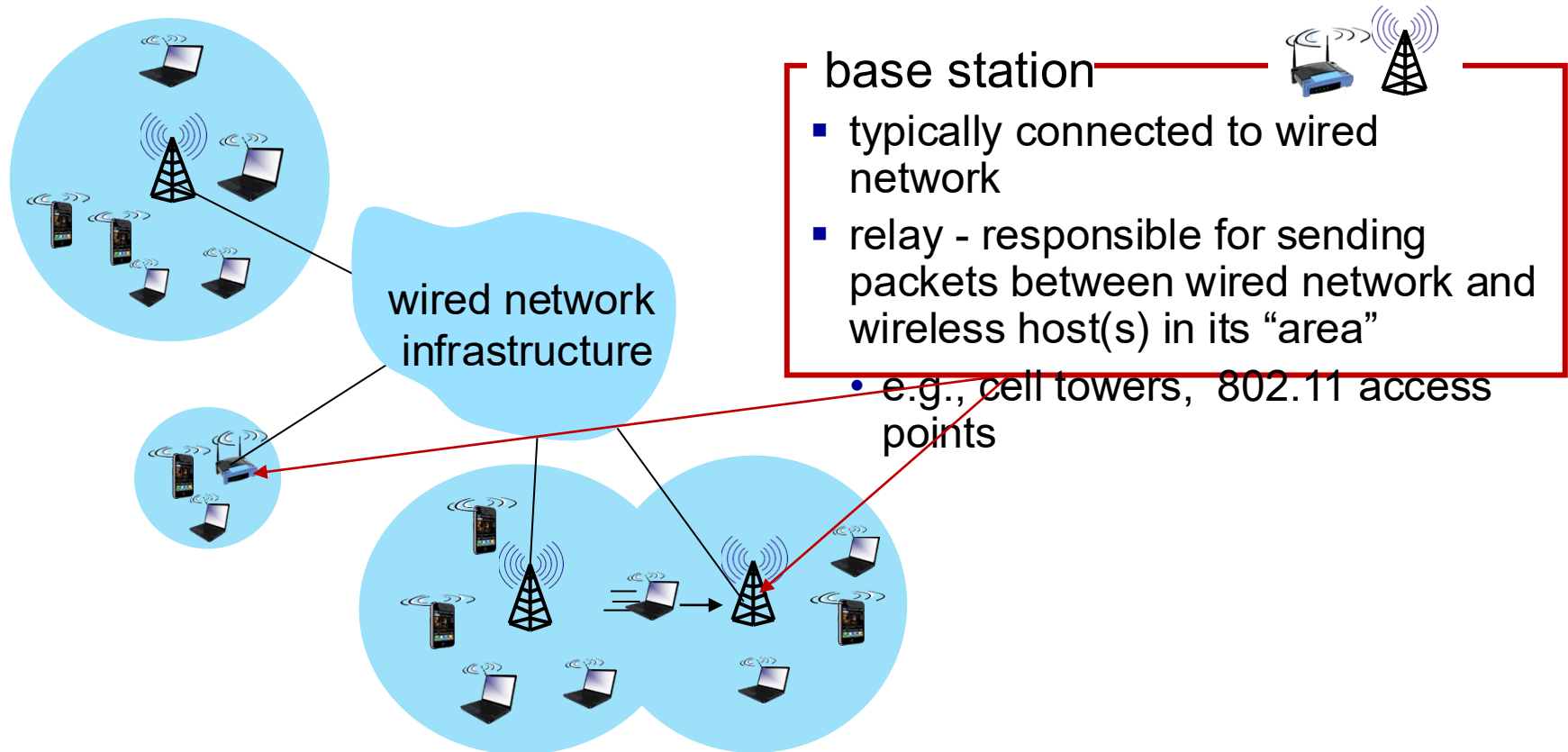
Elements of a wireless network



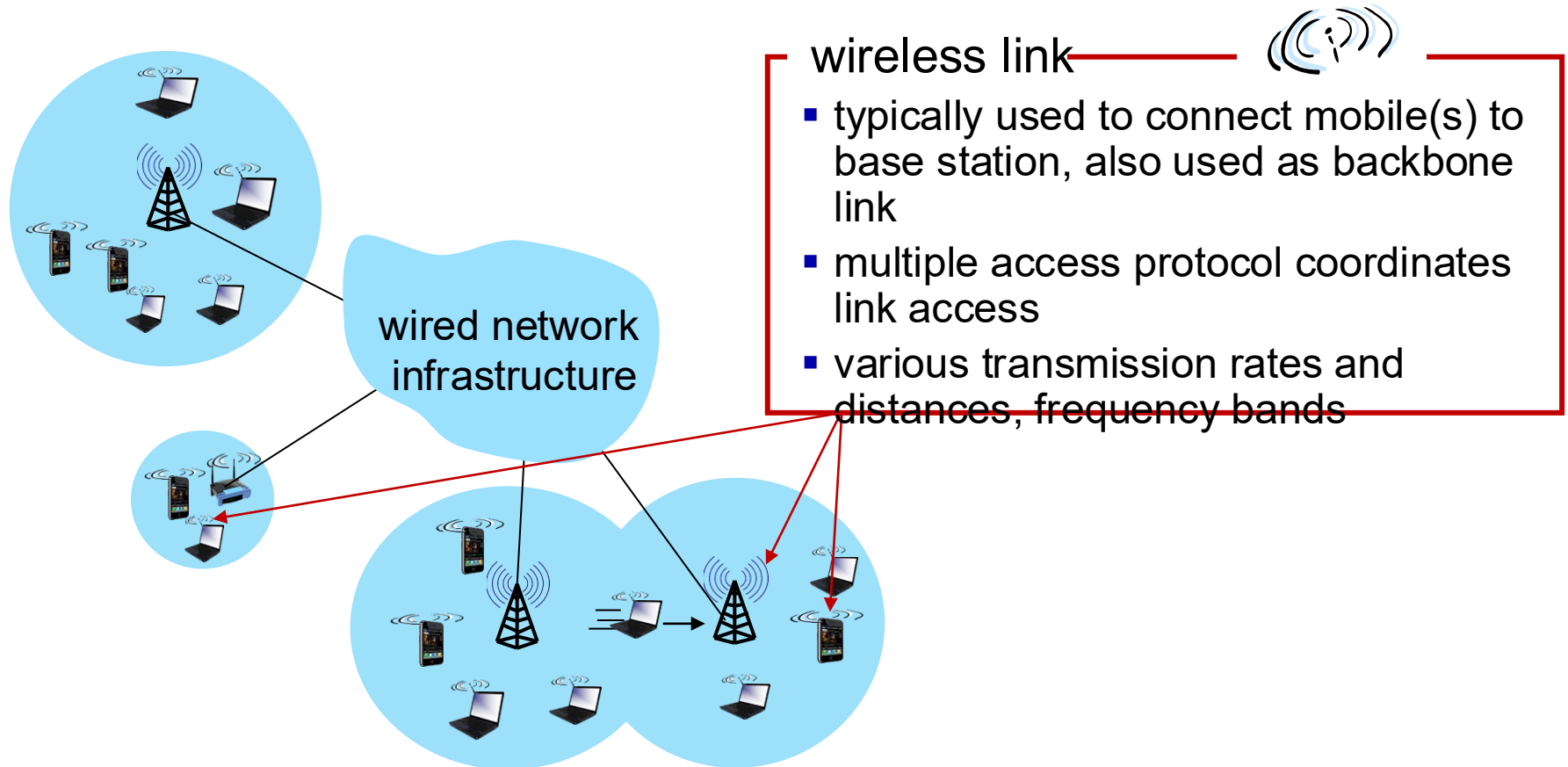
Elements of a wireless network



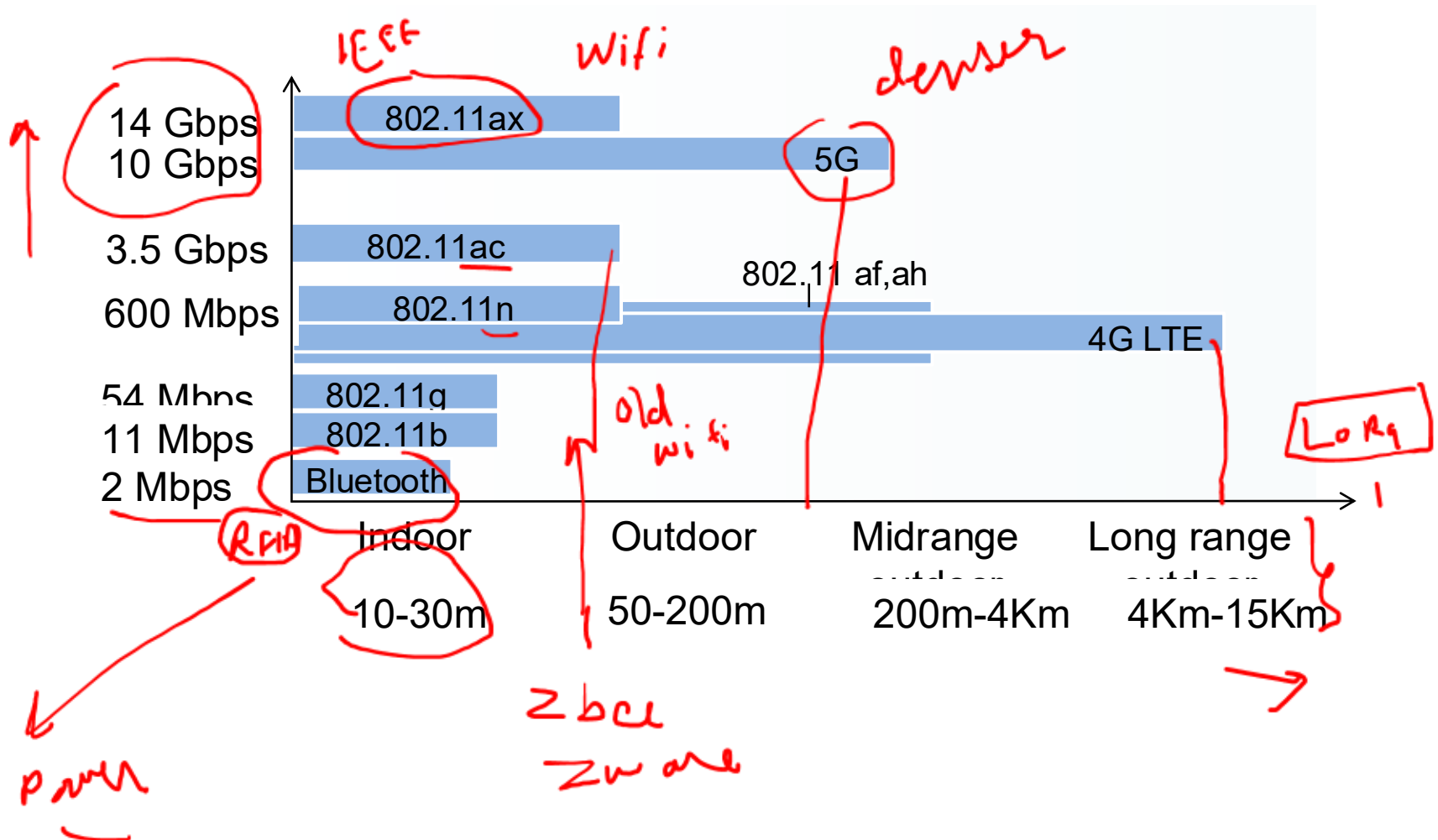
Elements of a wireless network



Elements of a wireless network

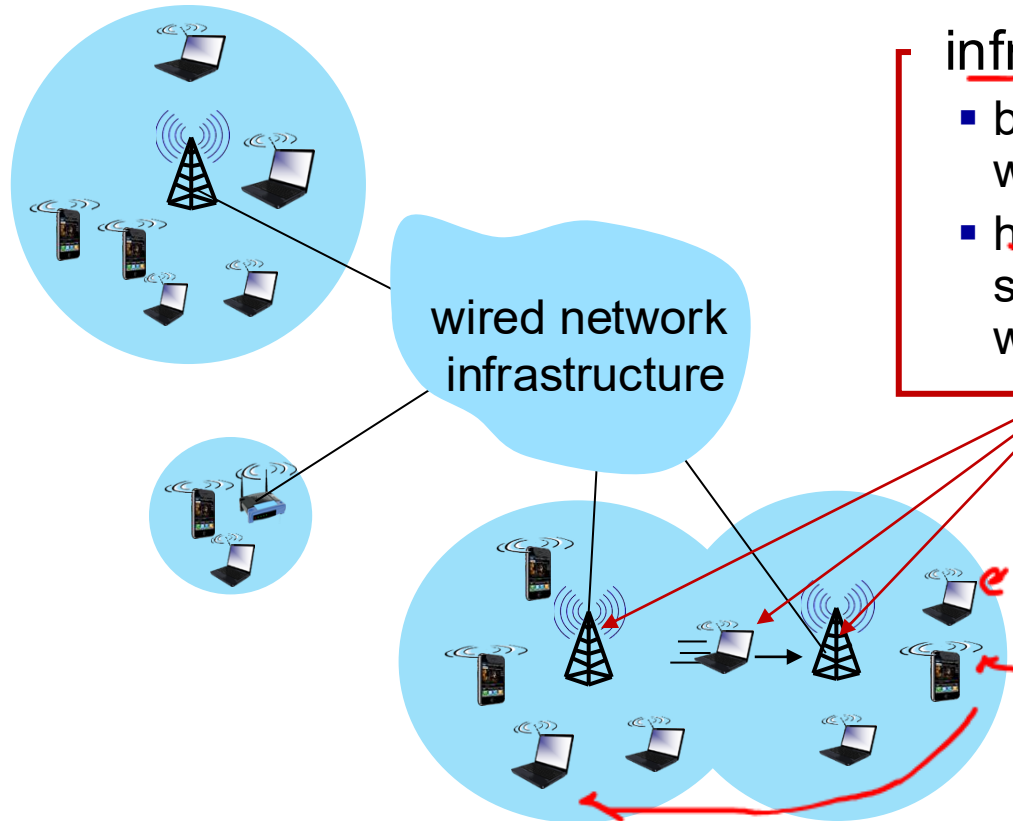


Characteristics of selected wireless links



Elements of a wireless network

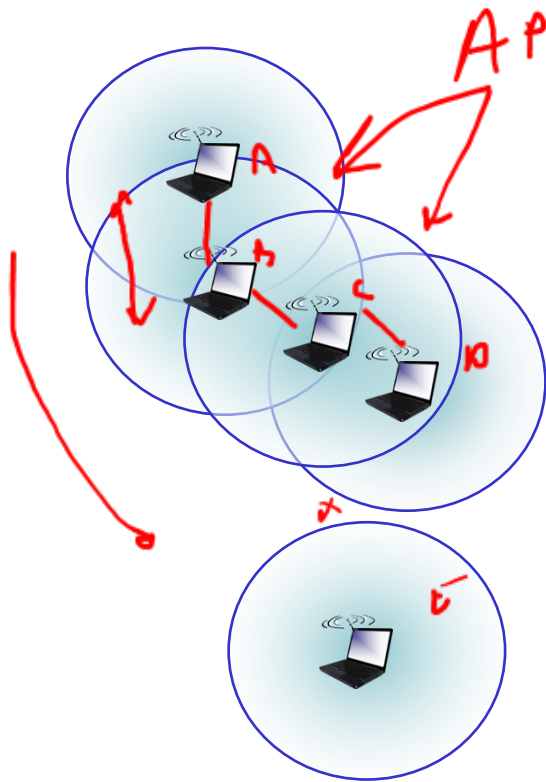
49/5 G



infrastructure mode

- base station connects mobiles into wired network
- handoff: mobile changes base station providing connection into wired network

Elements of a wireless network



ad hoc mode

wifi, bluetooth BLE

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

A → D

Wireless network taxonomy

A → D

A → B

	<u>single hop</u>	<u>multiple hops</u>
infrastructure (e.g., APs)	host connects to <u>base station</u> (WiFi, <u>cellular</u>) which connects to larger Internet	host may have to relay through several wireless nodes to connect to larger Internet: <u>mesh net</u>
<i>no infrastructure</i>	no base station, no connection to larger Internet (<u>Bluetooth</u> , ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given <u>wireless node</u> <u>MANET</u> , <u>VANET</u>

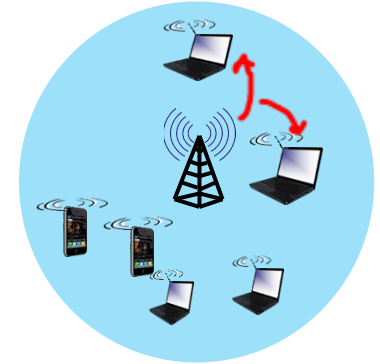
ad hoc

HAM

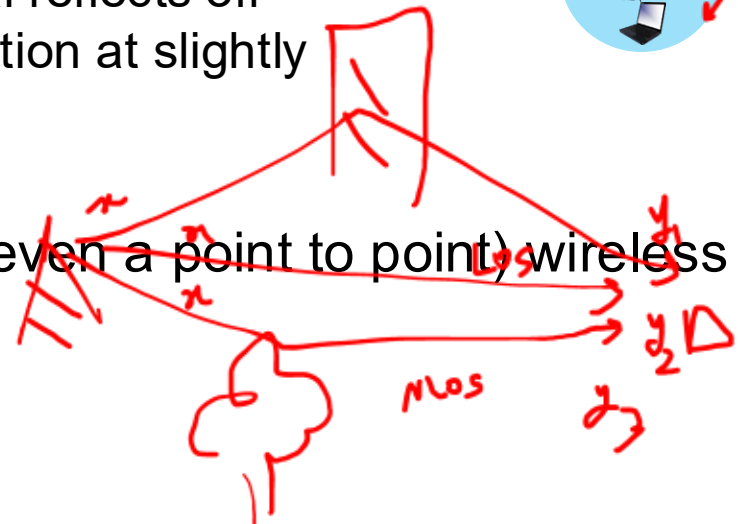
Wireless link characteristics (1)

important differences from wired link

- decreased signal strength: radio signal attenuates as it propagates through matter (path loss)
- interference from other sources: wireless network frequencies (e.g., 2.4 GHz) shared by many devices (e.g., WiFi, cellular, motors): interference
- multipath propagation: radio signal reflects off objects ground, arriving at destination at slightly different times



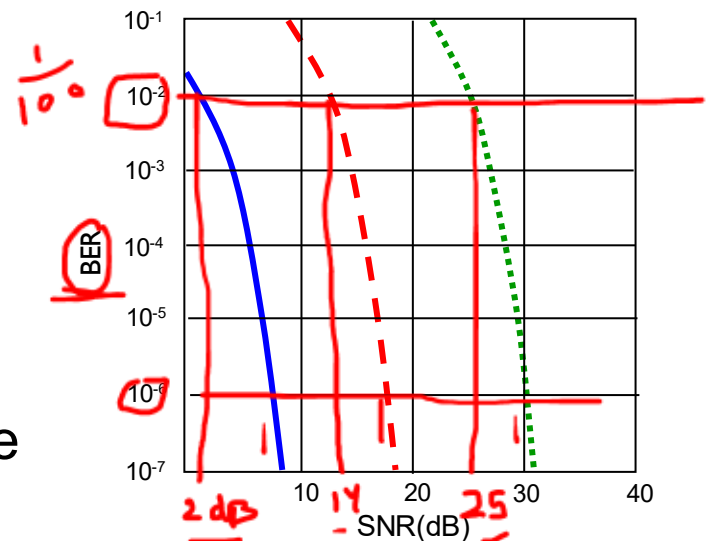
.... make communication across (even a point to point) wireless link much more "difficult"



$L_0 R_0$

Wireless link characteristics (2)

- SNR: signal-to-noise ratio
 - larger SNR – easier to extract signal from noise (a “good thing”)
- SNR versus BER (Bit Error Rate) tradeoffs
 - given physical layer: increase power → increase SNR → decrease BER
 - given SNR: choose physical layer that meets BER requirement, giving highest throughput
 - SNR may change with mobility: dynamically adapt physical layer (modulation technique, rate)



$\left. \begin{array}{l} \text{---} \text{QAM256 (8 Mbps)} \\ \text{---} \text{QAM16 (4 Mbps)} \\ \text{---} \text{BPSK (1 Mbps)} \end{array} \right\}$

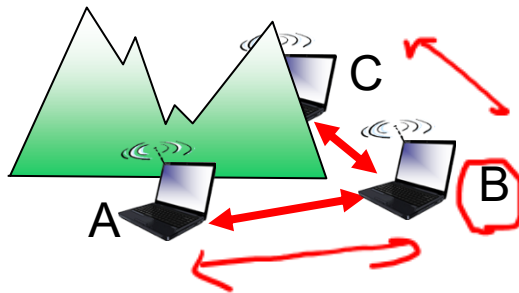
2 bits Modulation
Per symbol

Scheme

PHY + Link

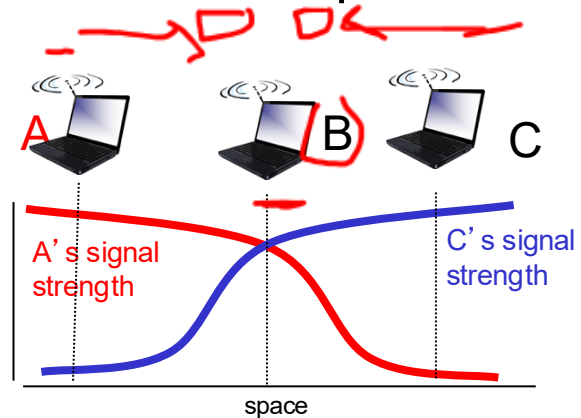
Wireless link characteristics (3)

Multiple wireless senders, receivers create additional problems (beyond multiple access):



Hidden terminal problem

- B, A hear each other
- B, C hear each other
- A, C can not hear each other means A, C unaware of their interference at B



Signal attenuation:

- B, A hear each other
- B, C hear each other
- A, C can not hear each other interfering at B

ACK / NACK

Antenna

- An antenna is an electrical conductor or system of conductors
 - Transmission – electrical energy converted to radiated electromagnetic energy into space
 - Reception - collects electromagnetic energy from space and converts into electrical
- In two-way communication, the same antenna can be used for transmission and reception
 - At the same time: Full-duplex
 - At different times: Half-duplex



Radiation



■ Radiation pattern

- An antenna will radiate power in all directions
- Typically, it does not radiate equally in all directions
- The distance from the antenna to each point on the radiation pattern is proportional to the power radiated by the antenna in that direction

■ An 'isotropic antenna' "doesn't exist"

- is an idealized antenna, represented by a point in space that radiates power in all directions equally
- Its radiation pattern is a sphere

■ Beam width (or half-power beam width)

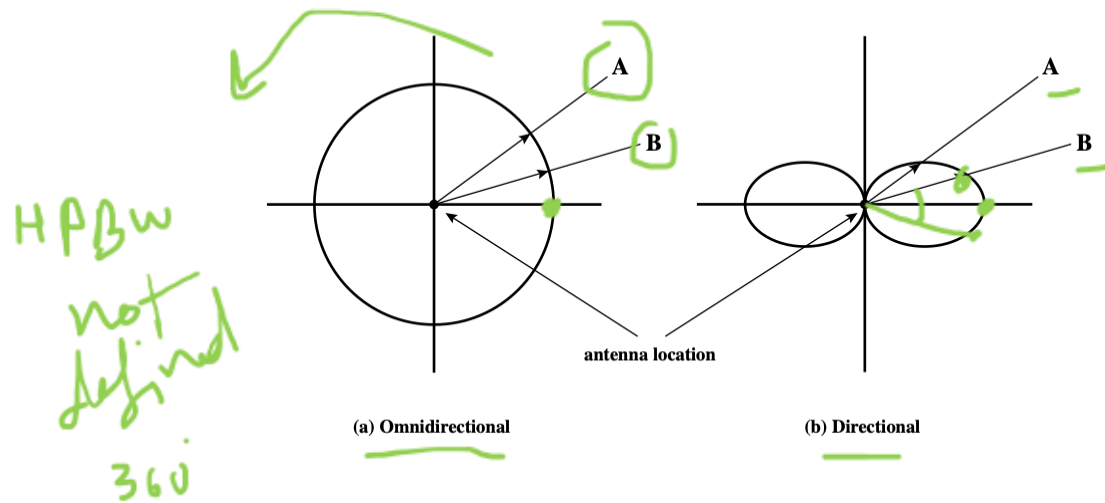
- Measure of directivity of antenna HPBW

■ Reception pattern

- ✓ Receiving antenna's equivalent to radiation pattern



Isotropic vs Anisotropic antennas



- Isotropic: omni-directional
- Anisotropic: directional

Types of Antennas

- Isotropic antenna (idealized)
 - Radiates power equally in all directions
- Wire antennas
 - Dipoles and Vertical Antennas
 - Loop antennas
 - Helix antennas

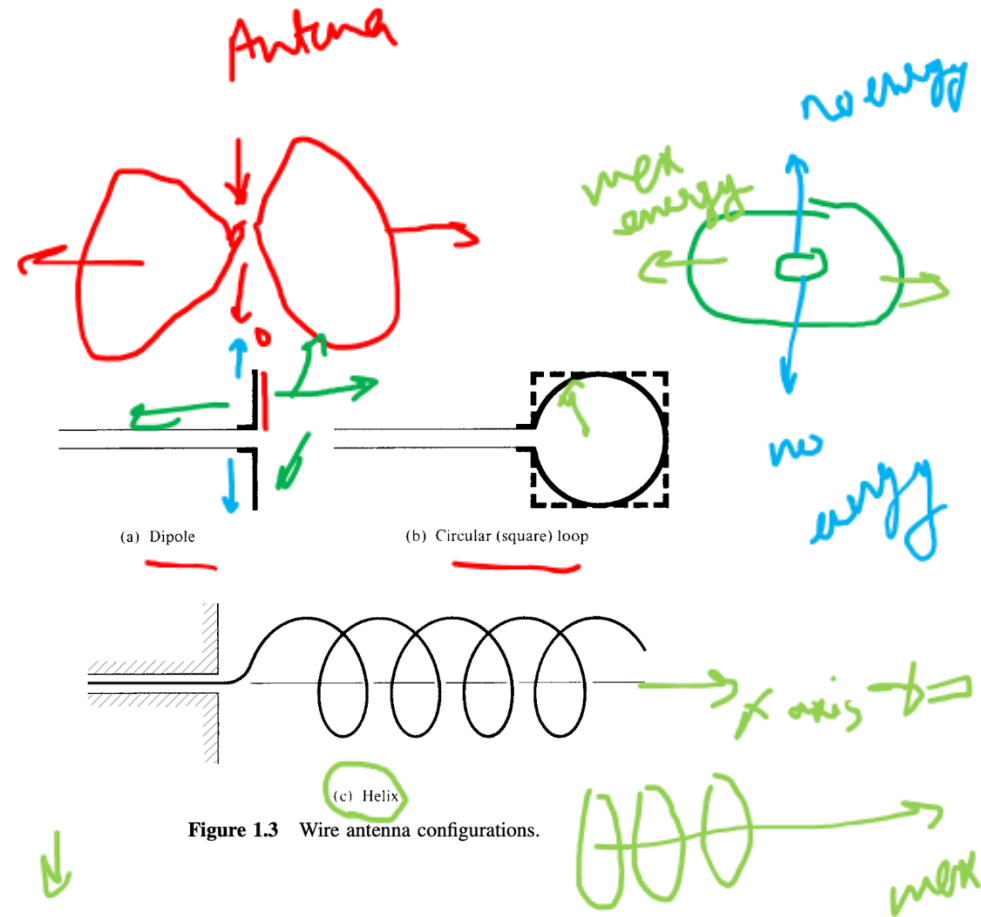


Figure 1.3 Wire antenna configurations.

[Src: Chapter 1, Balanis]

Types of Antennas (contd.)

- Exact wire shape matters
- Simple wire antennas
 - Dipole antennas
 - Half-wave dipole antenna (or Hertz antenna)
 - Vertical antennas
 - Quarter-wave vertical antenna (or Marconi antenna)

$$\lambda = \frac{c}{f}$$

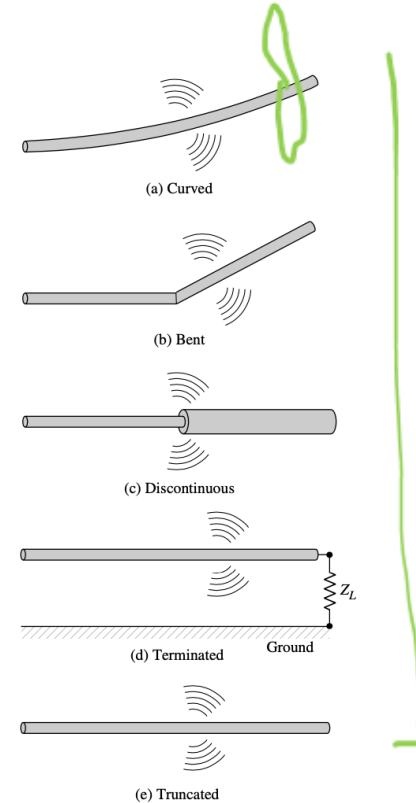
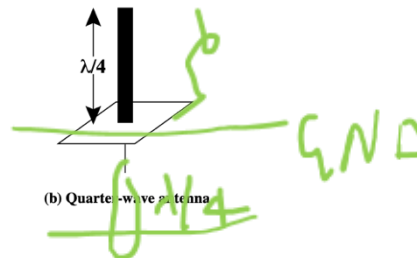
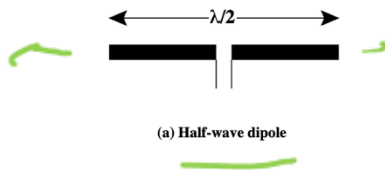


Figure 1.10 Wire configurations for radiation.

[Src: Chapter 1, Balanis]

Types of Antennas (contd.)

- Aperture antennas
 - Horn antennas
 - Conical antennas
 - Waveguides

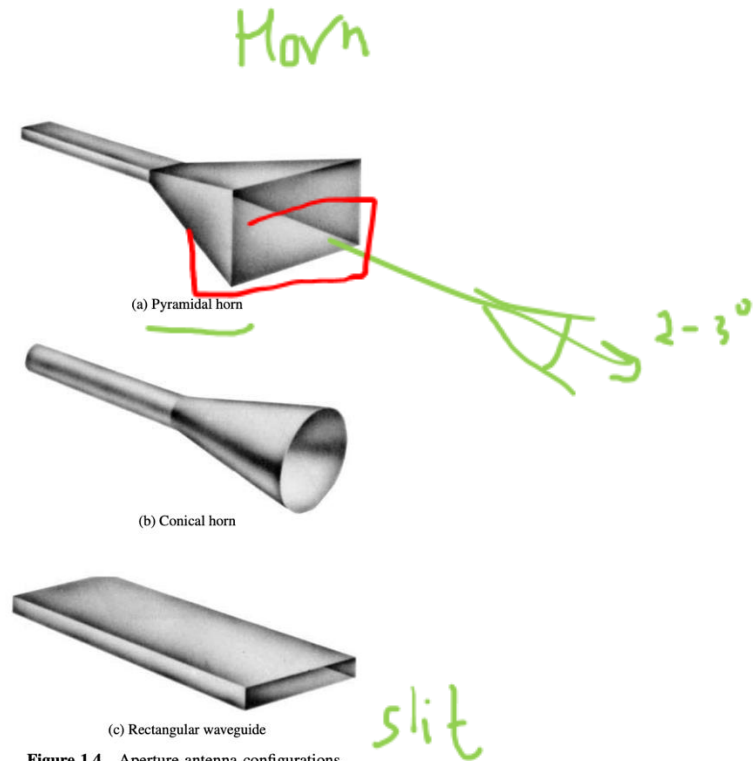


Figure 1.4 Aperture antenna configurations.

[Src: Chapter 1, Balanis]

Types of Antennas (contd.)

■ Microstrip or patch antennas

■ Ideal for small devices

- Small footprint
- Printed into ICs
- Easy to make arrays of

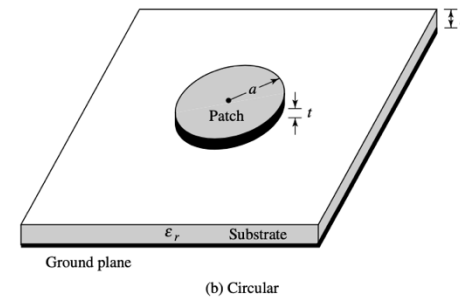
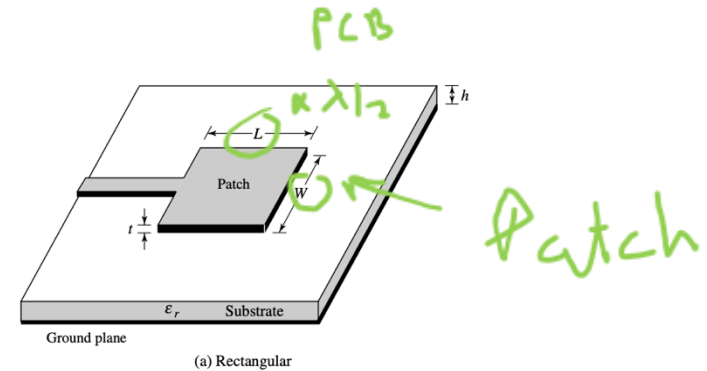
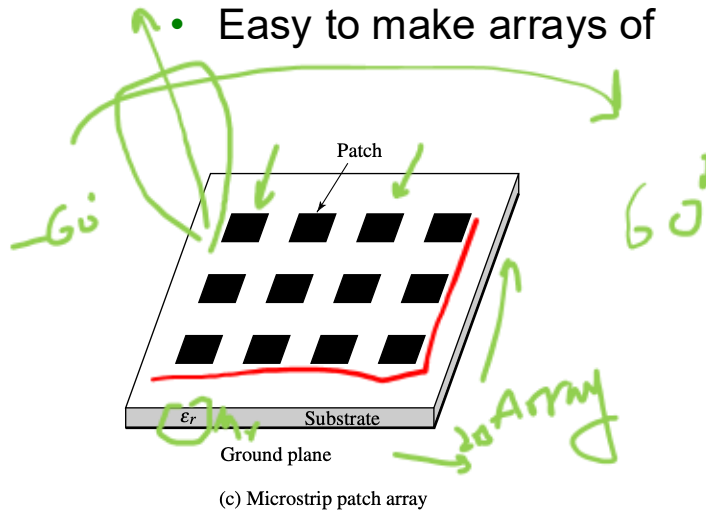


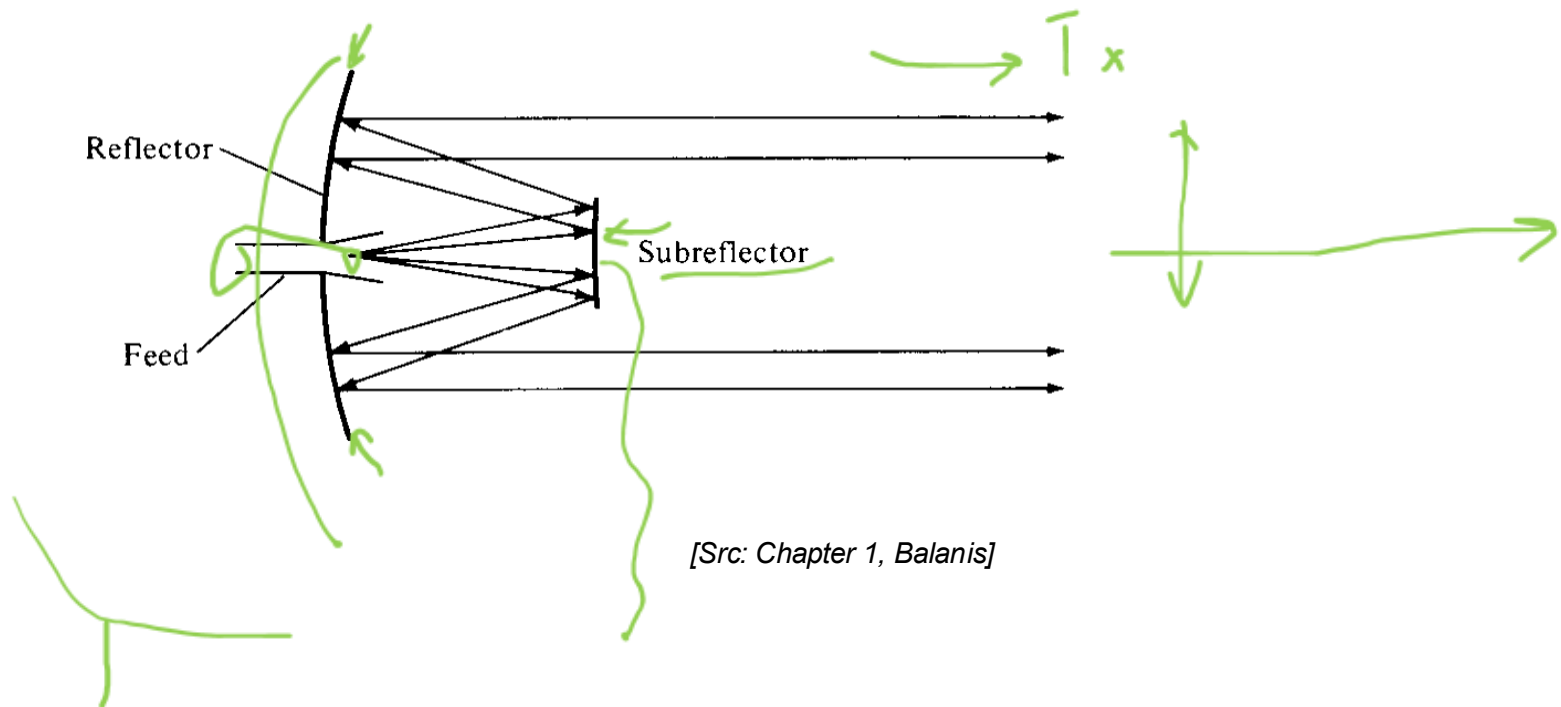
Figure 1.5 Rectangular and circular microstrip (patch) antennas.

[Src: Chapter 1, Balanis]

High

Types of Antennas (contd.)

- Lens Antennas
 - Also called parabolic reflective antenna



Antenna Gain

- Antenna gain:

- Power output, in a particular direction, compared to that produced in any direction by a perfect omnidirectional antenna (isotropic antenna) Gain = 0
- The increased power in a given direction is at the expense of other directions
- Refers to directionality, i.e. relative amplification in a certain direction

- Effective area Aperture

- Related to physical size and shape of antenna

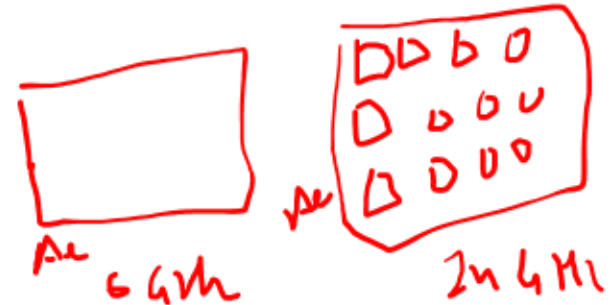
Antenna Gain (Cont'd)

- Relationship between antenna gain and effective area

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

Handwritten red annotations: A box around G , an arrow pointing from the box to the first equation, and two upward arrows labeled f and G to the right of the equations.

- G = antenna gain
- A_e = effective area
- f = carrier frequency
- c = speed of light ($= 3 \times 10^8$ m/s)
- λ = carrier wavelength
- Note: $\lambda f = c$



Antenna Gains and Effective Areas

TABLE 8-4 ANTENNA GAINS AND EFFECTIVE AREAS

Type of Antenna	Power Gain, G_A (absolute units)	Effective Area, A_e (m ²)
Isotropic	1	$\lambda^2/4\pi$
Infinitesimal dipole or loop	1.5	$1.5\lambda^2/4\pi$
Half-wave dipole	1.64	$1.64\lambda^2/4\pi$
Horn (optimized), mouth area, A	$10A/\lambda^2$	$0.81A$
Parabola or “dish” with face area, A	$7.0A/\lambda^2$	$0.56A$
Turnstile (two crossed dipoles fed 90° out of phase)	1.15	$1.15\lambda^2/4\pi$

[Src: Chapter 8, Couch]

Example

- Q: A parabolic reflective antenna with a diameter of 2m operating at 12 GHz. Find the antenna gain.

- G = antenna gain
- A_e = effective area
- f = carrier frequency
- c = speed of light ($= 3 \times 10^8$ m/s)
- λ = carrier wavelength

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

TABLE 8-4 ANTENNA GAINS AND EFFECTIVE AREAS

Type of Antenna	Power Gain, G_A (absolute units)	Effective Area, A_e (m ²)
Isotropic	1	$\lambda^2/4\pi$
Infinitesimal dipole or loop	1.5	$1.5\lambda^2/4\pi$
Half-wave dipole	1.64	$1.64\lambda^2/4\pi$
Horn (optimized), mouth area, A	$10A/\lambda^2$	$0.81A$
Parabola or “dish” with face area, A	$7.0A/\lambda^2$	$0.56A$
Turnstile (two crossed dipoles fed 90° out of phase)	1.15	$1.15\lambda^2/4\pi$